

JEE MAIN REVISION MASTER: SEMICONDUCTOR SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 92 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Logic Gates & Truth Tables	50 PYQs	Digital Electronics
Very High	p-n Junction Diode & Zener Diode Regulation	44 PYQs	Semiconductor Diodes

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Logic Gates & Truth Tables	$\begin{aligned} \text{NAND: } Y &= \overline{A \cdot B}, \quad \text{NOR: } Y = \overline{A + B}, \quad \overline{\overline{A \cdot B}} = \overline{\overline{A}} + \overline{\overline{B}}, \quad \overline{\overline{A + B}} = \overline{\overline{A}} \cdot \overline{\overline{B}} \end{aligned}$
p-n Junction Diode & Zener Diode Regulation	$\begin{aligned} \text{Efficiency}_{\text{half-wave}} &= 40.6\%, \quad \text{Efficiency}_{\text{full-wave}} = 81.2\%, \quad V_L = V_Z = \text{const}, \quad I_S = I_Z + I_L \end{aligned}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Logic Gates & Truth Tables (Very High Yield — 50 PYQs)

Core Principle: Universal logic gates (NAND and NOR), De Morgan's theorems, and Boolean circuit reduction.

■ **Caution / JEE Pitfall:** NAND and NOR gates are UNIVERSAL gates: any Boolean function can be constructed using only NAND or only NOR gates.

■ **Speed Trick:** Connect inputs of a NAND or NOR gate together to obtain a NOT gate: $\overline{A} \cdot A = \overline{A}$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "The output Y for the given inputs A and B to the circuit is :..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "For the given logic circuit, which of the following inputs combination will make both LED-1 and LED-2 to glow?..." [Ans: C]
- [2026 Evening Shift] Q#1 (MCQ): "If X and Y are the inputs, the given circuit works as _____." [Ans: D]

3.2 p-n Junction Diode & Zener Diode Regulation (Very High Yield — 44 PYQs)

Core Principle: Forward and reverse characteristics of p-n junctions, rectifiers, ripple frequency, and Zener diode shunt voltage stabilization.

■ **Caution / JEE Pitfall:** In full-wave rectifiers, output ripple frequency is TWICE the AC input frequency ($2f_{in}$), whereas for half-wave it is f_{in} .

■ **Speed Trick:** In reverse breakdown, Zener diode maintains constant voltage V_Z across its terminals as long as $I_Z \geq I_{Z,min}$.

Representative Source PYQs:

- **[2026 Morning Shift] Q#1 (MCQ):** "Find the correct combination of A,B,C and D inputs which can cause the LED to glow...." **[Ans: D]**
- **[2026 Evening Shift] Q#2 (MCQ):** "Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R Assertion A : A diode under reverse-b..." **[Ans: C]**
- **[2026 Morning Shift] Q#2 (MCQ):** "For the given circuit (shown in part (A)) the time dependent input voltage $v_{in}(t)$ and corresponding output $v_o(t)$ are shown in part..." **[Ans: B]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 5th April Morning Shift (Q#1)

For the given logic circuit, which of the following inputs combination will make both LED-1 and LED-2 to glow?

- (A) A=0,B=1,C=1
- (B) A=1,B=0,C=0
- (C) A=1,B=0,C=1 **[Correct]**
- (D) A=1,B=1,C=0

Correct Answer: C

Solution Analysis: Y_1 is the output of an OR gate whose inputs are A and B, $Y_1 = A + B$ Y_2 is an AND gate taking Y_1 and C as input $Y_2 = (A + B) \cdot C$ Y_3 is an AND gate taking Y_2 and A as input, $Y_3 = [(A + B) \cdot C] \cdot A$ $Y_3 = A \cdot (A + B) \cdot C = A \cdot C$ So, the truth table is, A B C $Y_1 = A + B$ $Y_2 = (A + B) \cdot C$ $Y_3 = A \cdot C$

A	B	C	$Y_1 = A + B$	$Y_2 = (A + B) \cdot C$	$Y_3 = A \cdot C$
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	1	0	0
0	1	1	1	1	0
1	0	0	1	0	0
1	0	1	1	1	1
1	1	0	1	0	1
1	1	1	1	1	1

LED1 glows when $Y_1 = 1$ and LED2 glows when $Y_3 = 1$ So for both the LEDs to glow we need $Y_1 = 1$ and $Y_3 = 1$

Benchmark Problem 2: JEE Main 2026 (Online) 22nd January Morning Shift (Q#1)

Find the correct combination of A,B,C and D inputs which can cause the LED to glow.

- (A) 1000
- (B) 0011
- (C) 0100
- (D) 1101 **[Correct]**

Correct Answer: D

Solution Analysis: Gate 1 is a NOR gate with both inputs tied together to A. A NOR gate with shorted inputs acts as a NOT gate. $Y_1 = \overline{A + A} = \overline{A}$ Inputs Output, Y 1 = A+ A = A 1 1 0 0 0 1 Gate 2 is also a NOR gate with both inputs tied to B, which acts as a NOT gate. $Y_2 = \overline{B + B} = \overline{B}$ Inputs Output, Y 2 = B+ B = B 1 1 0 0 0 1 Gate 3 is a NAND gate receiving inputs Y_1 and Y_2 . $Y_3 = \overline{Y_1 \cdot Y_2}$ A B Y 1 = A Y 2 = B Y 1 · Y 2 $Y_3 = \overline{Y_1 \cdot Y_2}$ 1 1 0 0 0 1 1 0 0 1 0 1 0 1 0 1 1 0 0 1 0 0 1 1 1 0 Gate 4 is a NOR gate receiving inputs C and D. $Y_4 = \overline{C + D}$ C D C